Zhanwei Zhang (张展玮)

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Education

M.Phil. in Data Science and Analytics, Hong Kong University of Science and Technology (Guangzhou) **Sep 2025 ~ Present**
Information Hub
Advisor: Prof. Zishuo Ding

B.Sc. in Computer Science and Technology, Southern University of Science and Technology (SUSTech) **Sep 2021 ~ Jun 2025**
Turing Master Class
Advisor: Prof. Yepang Liu
GPA: 3.79 / 4.0 | Weight Avg Score: 90.92 | Ranking: 36 / 195
Main courses: Mathematical Logic (A+), Java Programming (A+), Data Structures & Algorithms (A), Machine Learning (A), Database Systems (A-), Compilers (B+)

Internships

Agent Platform Intern in Qualcomm, Shenzhen **May 2026 ~ Present**
Core developer of an internal AI Agent platform, evolving its architecture and enabling unified Agent / Workflow / Tool integration.
Designed Engine + Plugin architecture with dynamic backend/frontend loading via Python Entry Points and Module Federation.
Built Agent Runtime on DeepAgents, unifying Tool / MCP / Skills / Subagent integration and supporting AG-UI and HITL.

Large Language Model Intern in Lingsome, Shenzhen **Aug 2024 ~ Aug 2025**
Integrated multi-type Retrieval Augmented Generation (RAG) and GraphRAG systems.
Optimized GraphRAG code to extract better entities & relationships and construct domain-specific knowledge graphs.
Developed and refined pipelines to extract improved entities and relationships.

Visiting Researcher in Wuhan University, Wuhan **May ~ Aug 2024**
Advisor: Prof. Jinfu Chen (WHU); Prof. Weiyi Shang (UWaterloo)
Focused on software logging and failure workarounds.
Developed an automated analysis pipeline to extract, filter, and sample code commits containing try-catch blocks.

Selected Projects

OpenStar **Jul 2026 ~ Present**
Built Chat / Explore / Watch Agent workflows for open-source project discovery, understanding, and continuous tracking.
Designed Agent research and Watch pipelines to investigate README / Release / Activity via tools and produce structured assessments.

WiTH - AI Health Companion **Jun 2026**
Built an AI health Agent integrating user profiles, health records, and 20+ Tool Calling capabilities for personalized health interactions.
Implemented semantic memory and proactive health reminders, turning conversations into actionable tasks; won Third Prize at the Shenzhen Hackathon.

AI Micro-Drama Studio **Sep 2025 ~ Jan 2026**
Built an end-to-end pipeline that transforms novels or scripts into production-ready storyboards.
Unified creation, generation, editing, and export in one interface to streamline publish-ready video production.

Customizable AI Companion Doll (OpenHarmony TSC Project) **Sep 2025 ~ Jan 2026**
Developed an AIoT system integrating cloud-based LLMs to provide personalized companionship through customizable personalities and multimodal interaction.
Built the cross-platform frontend, including a mobile App and Mini-program.

Othello Game through Java and Python Programming with Strong AI **Oct ~ Dec 2021 & Mar 2023**

Developed visually appealing interface and implemented Monte Carlo & Alpha Beta Pruning algorithm.

Canteen Traffic Monitoring **Dec 2023 ~ Jan 2024**

Calculated the length of the queue by monitoring data and displayed a chart showing the changes in queue length.

Won award for finalist in National College Students' Innovation and Entrepreneurship Training program.

About 30,000 visits within three months.

Simple Compiler **Sep 2023 ~ Jan 2024**

Developed a compiler that translates C language files into Intermediate Representation (IR) and MIPS32.

Supported essential features such as I/O operations, control flow and function calls.

Research

Numerical Error Detection in Floating-Point Computing **Sep 2024 ~ Feb 2026**

PI-detector: A condition-number-guided perturbation approach can replace costly high-precision oracles, finding 173/174 significant-error cases at about 0.13% of oracle cost (up to 73.46× faster).

MGDE: Turn detection into a Newton-Raphson-guided convergent input search, reaching 80 bugs / 47 functions (vs. 70/46 ATOMU, 53/42 FPCC) while being 41.71× and 10.17× faster, respectively.

Reproduction and Evaluation of R1-style Reasoning Pipeline **May 2025 ~ Jun 2025**

Replicated the DeepSeek-R1 training pipeline by implementing GRPO and Cold-start SFT, significantly enhancing the multi-step reasoning capabilities of Qwen2.5 models (0.5B to 7B).

LLM-Based JSON Parser Fuzzing for Bug Discovery and Behavioral Analysis **Sep 2023 ~ Jan 2024**

Used opensource LLMs such as Llama2-7B/13B to generate test cases.

13 JSON Parsers and over 100 types of cases have been tested. Over 26 behavioral diversities have been found.

Publications

Tan, Y., **Zhang, Z.**, Ding, Z., Chen, J., & Shang, W. (2026). *ICE: Reducing search space for error-inducing input detection. IEEE Transactions on Software Engineering*, 1-14. doi:10.1109/TSE.2026.3703362

Tan, Y., **Zhang, Z.**, Zhang, H., Zheng, L., Ding, Z., Chen, J., & Shang, W. (2026). *Mathematically-guided detection of floating-point errors. ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA 2026)*. ResearchGate (Accepted)

Tan, Y., **Zhang, Z.**, Chen, J., Ding, Z., Xuan, J., & Shang, W. (2025). *Computing floating-point errors by injecting perturbations.* arXiv. https://arxiv.org/abs/2507.08467

Han, Y., Shen, H., He, X., Mai, Z., Zhang, R., Zheng, Z., Liu, Y., Zhang, X., Li, G., **Zhang, Z.**, Liang, Z., Chen, Y., Xie, Y., Li, M., Shen, G., Wang, C., Ye, J., Zhu, L., Fu, T.-M., & Yang, X. (2025). A comprehensive analysis of interflight variability in carbon dioxide emissions from global aviation. *Environmental Science & Technology*, 59(12), 6179-6191. doi:10.1021/acs.est.5c02371

Patents

一种点餐方法、系统、终端及介质 (Innovative Ordering Method, System, Terminal, and Medium Patent) **May 2023**

Innovated a method and system to alleviate peak-hour traffic in cafeterias.

Applied on May 5, 2023; Application no: 202310498065

Skills

Languages: English (Fluent; IELTS: 6.5), Mandarin (Native), Cantonese (Native)

Programming Languages & Frameworks: Java, Python, C/C++, SQL, Spring Boot, Vue, React

Tools: IntelliJ IDEA, PyCharm, Visual Studio Code, Anaconda, Git, CMake

Honors & Scholarships

Third Prize, Shenzhen Hackathon **Jun 2026**

Special Innovation Award (Unique Winner) & Second Prize, OpenHarmony Competition Training Camp **Sep 2025**

Postgraduate Studentship (PGS), HKUST(GZ)	**Sep 2025**
Outstanding Student, SUSTech	**Jan 2024**
Honorable Mention, Mathematical Contest in Modeling	**May 2023**
Finalist, National College Students' Innovation and Entrepreneurship Training program	**Jun 2023**
Third Prize, China Undergraduate Mathematical Contest in Model	**Sep 2023**